

PSPJ - PROJECT

MODULE 1

TITLE: ADAPTIVE QUIZ AND EXAM ENGINE

Abstract

The **Adaptive Quiz & Exam Engine** is a simple quiz system that changes the difficulty of questions based on how well a student is performing. The quiz starts with an easy question. If the student answers correctly, the difficulty increases, and if the answer is wrong, the difficulty decreases. This allows the quiz to adjust to the student's performance instead of giving the same level of questions throughout the test.

This project helped us apply the main ideas of **computational thinking**, such as decomposition, pattern recognition, abstraction, and algorithm design. We first broke the problem into smaller steps and identified the inputs, outputs, assumptions, constraints, and possible edge cases. We also used pseudocode, a flowchart, and a trace table to understand and test the logic before writing the actual program.

The system is developed using **Java** and uses basic concepts such as variables, data types, constants, operators, **if-else** statements, loops, type conversion, and **Scanner** for user input. At the end of the quiz, the program calculates the student's score and percentage and displays the final performance.

Through this project, we learned how a real-life problem can be broken down into clear steps, converted into an algorithm, and finally implemented as a working Java program.

Advantages

Advantages: The Adaptive Quiz & Exam Engine provides immediate feedback, automatically calculates scores, adjusts question difficulty according to performance, reduces manual work, and demonstrates important Java and computational-thinking concepts. It is also simple to understand, easy to modify, and can be expanded with more questions and features in the future.

IMPLEMENTATION:

```
import java.util.Scanner;
```

```
class QuizEngine {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        System.out.println("=====");

        System.out.println("    ADAPTIVE QUIZ & EXAM ENGINE");

        System.out.println("=====");

        System.out.print("Enter your name: ");

        String studentName = input.nextLine();

        final int TOTAL_QUESTIONS = 5;

        final double PASS_MARK = 50.0;

        int score = 0;

        int correctAnswers = 0;

        int incorrectAnswers = 0;

        int questionNumber = 1;

        // 1 = Easy

        // 2 = Medium

        // 3 = Hard

        int difficulty = 1;

        boolean quizStarted = true;

        while (questionNumber <= TOTAL_QUESTIONS) {
```

```
System.out.println();

System.out.println("-----");

System.out.println("Question " + questionNumber);
```

```
// -----
```

```
// DISPLAY DIFFICULTY
```

```
// -----
```

```
if (difficulty == 1) {

    System.out.println("Difficulty: EASY");

}

else if (difficulty == 2) {

    System.out.println("Difficulty: MEDIUM");

}

else {

    System.out.println("Difficulty: HARD");

}
```

```
// -----
```

```
// QUESTION 1
```

```
// -----
```

```
if (questionNumber == 1) {

    System.out.println("What is 5 + 5?");

    System.out.println("A. 10");

    System.out.println("B. 15");

    System.out.println("C. 20");

    System.out.println("D. 25");

}
```

```
System.out.print("Enter your answer: ");
```

```
char answer = input.next().toUpperCase().charAt(0);
```

```
char correctAnswer = 'A';
```

```
if (answer == correctAnswer) {
```

```
    System.out.println("Correct!");
```

```
    score++;
```

```
    correctAnswers++;
```

```
    // Correct → increase difficulty
```

```
    if (difficulty < 3) {
```

```
        difficulty++;
```

```
    }
```

```
}
```

```
else {
```

```
    System.out.println("Incorrect!");
```

```
    incorrectAnswers++;
```

```
    // Incorrect → decrease difficulty
```

```
    if (difficulty > 1) {
```

```
        difficulty--;
```

```
    }
```

```
}
```

```
}
```

```
// -----
```

```
// QUESTION 2
```

```
// -----
```

```
else if (questionNumber == 2) {
```

```
    System.out.println("What is 10 * 2?");
```

```
    System.out.println("A. 10");
```

```
    System.out.println("B. 20");
```

```
    System.out.println("C. 30");
```

```
    System.out.println("D. 40");
```

```
    System.out.print("Enter your answer: ");
```

```
    char answer = input.next().toUpperCase().charAt(0);
```

```
    char correctAnswer = 'B';
```

```
    if (answer == correctAnswer) {
```

```
        System.out.println("Correct!");
```

```
        score++;
```

```
        correctAnswers++;
```

```
        if (difficulty < 3) {
```

```
            difficulty++;
```

```
        }
```

```
    }
```

```
else {

    System.out.println("Incorrect!");

    incorrectAnswers++;

    if (difficulty > 1) {

        difficulty--;

    }

}

// -----

// QUESTION 3

// -----

else if (questionNumber == 3) {

    System.out.println(

        "Which data type stores whole numbers in Java?"

    );

    System.out.println("A. double");

    System.out.println("B. boolean");

    System.out.println("C. int");

    System.out.println("D. char");

    System.out.print("Enter your answer: ");

    char answer = input.next().toUpperCase().charAt(0);
```

```
char correctAnswer = 'C';

if (answer == correctAnswer) {

    System.out.println("Correct!");

    score++;

    correctAnswers++;

    if (difficulty < 3) {

        difficulty++;

    }

}

else {

    System.out.println("Incorrect!");

    incorrectAnswers++;

    if (difficulty > 1) {

        difficulty--;

    }

}

// -----

// QUESTION 4

// -----

else if (questionNumber == 4) {
```

```
System.out.println(

    "Which operator checks equality in Java?"

);

System.out.println("A. =");

System.out.println("B. ==");

System.out.println("C. !=");

System.out.println("D. >=");

System.out.print("Enter your answer: ");

char answer = input.next().toUpperCase().charAt(0);

char correctAnswer = 'B';

if (answer == correctAnswer) {

    System.out.println("Correct!");

    score++;

    correctAnswers++;

    if (difficulty < 3) {

        difficulty++;

    }

}

else {

    System.out.println("Incorrect!");

    incorrectAnswers++;

    if (difficulty > 1) {

        difficulty--;

    }

}

}
```

```

// -----

// QUESTION 5

// -----

else {

    System.out.println("What does JVM stand for?");

    System.out.println("A. Java Variable Machine");

    System.out.println("B. Java Virtual Machine");

    System.out.println("C. Java Verified Machine");

    System.out.println("D. Java Visual Machine");

    System.out.print("Enter your answer: ");

    char answer = input.next().toUpperCase().charAt(0);

    char correctAnswer = 'B';

    if (answer == correctAnswer) {

        System.out.println("Correct!");

        score++;

        correctAnswers++;

        if (difficulty < 3) {

            difficulty++;

        }

    }

    else {

        System.out.println("Incorrect!");

        incorrectAnswers++;

        if (difficulty > 1) {

            difficulty--;

        }

    }

}

// -----

// MOVE TO NEXT QUESTION

```

```

// -----

    questionNumber++;

}

// =====

// OUTPUT

// =====

double percentage =

    (score * 100.0) / TOTAL_QUESTIONS;

System.out.println();

System.out.println("=====");

System.out.println("        FINAL RESULT");

System.out.println("=====");

System.out.println("Student Name: " + studentName);

System.out.println("Total Questions: " + TOTAL_QUESTIONS);

System.out.println("Correct Answers: " + correctAnswers);

System.out.println("Incorrect Answers: " + incorrectAnswers);

System.out.println("Score: " + score);

System.out.println("Percentage: " + percentage + "%");

// =====

// FINAL DIFFICULTY

// =====

if (difficulty == 1) {

    System.out.println("Final Difficulty: EASY");

}

else if (difficulty == 2) {

    System.out.println("Final Difficulty: MEDIUM");

}

else {

    System.out.println("Final Difficulty: HARD");

}

```

```

// =====

// PASS / FAIL

// =====

boolean passed = percentage >= PASS_MARK;

if (passed && quizStarted) {

    System.out.println("Result: PASS");

}

else {

    System.out.println("Result: FAIL");

}

// =====

// TERNARY OPERATOR

// =====

String performance =

    percentage >= 80 ? "Excellent" :

    percentage >= 60 ? "Good" :

    percentage >= 50 ? "Average" :

    "Needs Improvement";

System.out.println("Performance: " + performance);

// =====

// CLOSE SCANNER

// =====

input.close();

System.out.println();

System.out.println("Thank you for using the system!!");

}

}

```

OUTPUT:

```
Difficulty: HARD
Which operator checks equality in Java?
A. =
B. ==
C. !=
D. >=
Enter your answer: B
Correct!

-----
Question 5
Difficulty: HARD
What does JVM stand for?
A. Java Variable Machine
B. Java Virtual Machine
C. Java Verified Machine
D. Java Visual Machine
Enter your answer: B
Correct!

=====
FINAL RESULT
=====
Student Name: BHAVYA
Total Questions: 5
Correct Answers: 5
Incorrect Answers: 0
Score: 5
Percentage: 100.0%
Final Difficulty: HARD
Result: PASS
Performance: Excellent

Thank you for using the system!!
PS C:\PSPJ\work\week-02> █
```

CONCLUSION:

Conclusion

Through this project, we learned how to take a real-world problem and convert it into a clear computational solution. The **Adaptive Quiz & Exam Engine** shows how a quiz can change its difficulty based on a student's answers. We used computational thinking concepts such as decomposition, pattern recognition, abstraction, and algorithm design to plan the system before writing the Java program.

While developing the program, we also applied basic Java concepts such as variables, data types, operators, **if-else** statements, loops, constants, type conversion, and **Scanner** for taking input. We used a flowchart, pseudocode, and trace table to understand and test the logic of the program.

Overall, this project(MODULE 1) helped us understand that programming is not only about writing code. It is also about understanding a problem, breaking it into smaller parts, creating a logical solution, testing it, and then converting that solution into a working program. The project also gave us a better understanding of how Java can be used to solve practical problems in a simple and structured way.

